

## 2024 AMC 10A Problem 4 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

### 2024 AMC 10A Problem 4

**Problem:**

The number 2024 is written as the sum of not necessarily distinct two-digit numbers. What is the least number of two-digit numbers needed to write this sum?

**Answer Choices:**

- A. 20
- B. 21
- C. 22
- D. 23
- E. 24

**Solution:**

In order to minimize the number of terms in the sum, the greatest two-digit number, 99, should be used as many times as possible. Because  $20 \cdot 99 = 1980$ , the minimum number of terms is greater than 20. On the other hand,  $2024 = 20 \cdot 99 + 44$ , so the least number of two-digit numbers needed is (B) 21.

*The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗*

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